TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Quantitative Phase Imaging Platform for Label Free CTC Analysis in Metastatic Breast Cancer**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medicine			
	Medium Category	Small Category	Medium Category	Small Category
	Treatment and diagnosis devices	Clinical chemistry and biology analytic apparatus		
Concept of Proposed Technology	We propose a breakthrough blood-based diagnostic platform for the early detection and monitoring of metastatic breast cancer. The technology leverages label-free quantitative phase imaging to identify and analyze circulating tumor cells directly from blood samples, without staining or molecular markers. By capturing the intrinsic optical and biophysical signatures of single cells and combining them with AI-driven analysis, the platform enables fast, non-invasive, and highly informative cancer assessment. It is designed to support early metastasis risk stratification, longitudinal disease monitoring, and therapy response evaluation, offering a powerful alternative to invasive tissue biopsies and opening new pathways for precision oncology in routine clinical practice.			
Rationale for Joint R&D Project	This joint R&D project addresses a clear unmet need for robust, label-free, and clinically scalable analysis of circulating tumor cells, which cannot be solved by a single partner alone. The challenge spans advanced optical metrology, quantitative image reconstruction, AI-based biomarker extraction,			

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	clinical validation, and regulatory readiness, requiring tightly coupled expertise across disciplines. Collaboration is essential because Fraunhofer IPM contributes unique know-how in digital holographic quantitative phase imaging, high-speed reconstruction algorithms, and measurement standardization, while the industrial partner provides clinically validated CTC isolation, access to patient samples, and pathways to commercialization. Only the integration of these complementary capabilities enables the translation of a laboratory-grade imaging concept into a clinically relevant, CE-ready diagnostic platform with validated performance and market viability.
Key Words	1. Digital Holographic Microscopy (DHM) 2. Label-free cytology 3. Circulating Tumor Cells (CTCs) 4. Breast cancer metastasis 5. Digital pathology 6. AI-driven biomarkers

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Development of a non-invasive, real-time diagnostic platform that uses digital holographic QPI to analyze circulating tumor cells from blood samples of breast cancer patients. The goal is to enable label-free, rapid detection and characterization of CTCs for predicting metastatic progression in breast cancer. This QPI-based approach will provide quantitative, marker-independent insights into tumor cell biology, supporting early metastasis risk stratification and guiding timely therapeutic decisions. By replacing invasive tissue biopsies with a blood test, the technology aims to improve patient monitoring and outcomes in metastatic breast cancer management.			
R&D Target Specifications	1. **Standardized QPI Dataset** a. ≥ **30,000 single-cell QPI images** (CTCs + background blood cells) 2. **Label-Free CTC Detection Performance** a. CTC vs WBC classification **accuracy ≥ 90%** b. Sensitivity ≥ **90%**, specificity ≥ **95%** 3. **QPI-Derived Digital Biomarkers** a. ≥ **5 quantitative biomarkers per cell** (e.g., dry mass, mean phase delay, refractive index distribution, optical thickness) b. Measurement reproducibility **CV ≤ 5%** 4. **Metastasis Prediction Capability** a. Longitudinal (pre/post-treatment) CTC analysis b. Metastasis prediction model **ROC-AUC ≥ 0.85** 5. **System Throughput and Speed** a. ≤ **3 minutes per sample** (CTC capture + QPI imaging) b. ≥ **1,000 cells/min** processing capacity 6. **Regulatory & Commercial Readiness** a. **TRL 6–7** achieved 7. ≥ **1 CE-IVDR–aligned technical documentation package** prepared			
R&D Plan	Fraunhofer IPM will contribute an advanced DHM/QPI optical system and imaging algorithm as the core of the platform. This system records digital holograms of cells and reconstructs quantitative phase images, capturing each cell's optical phase delay with sub-micrometer precision. The technique records the phase information of cells, yielding rich morphological and intracellular data without any staining. Fraunhofer's expertise in aberration correction and high-speed phase reconstruction (e.g., GPU-accelerated hologram processing at 100+ million data points per second) will ensure the imaging is fast and accurate enough for real-time analysis. The platform will be integrated in a circulating tumor cell isolation workflow, which uses a gravity-driven High-Density Micro-porous (HDM) chip to enrich intact, viable CTCs from blood with minimal cell damage. This integration allows label-free optical analysis immediately after CTC capture, maintaining cell integrity and avoiding complex preparation.			

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	In collaboration with a clinical team, the project will replace or augment the current immunofluorescence-based CTC identification step with QPI-based digital cytology. The QPI images provide an endogenous contrast via the cells' refractive index, eliminating reliance on surface markers and enabling detection of heterogeneous CTC populations. Co-development of novel digital biomarkers is a key focus: from each cell's phase image, quantitative parameters such as optical phase shift, cell dry mass, volume, and refractive index distribution will be extracted. These label-free QPI biomarkers reflect the cell's biophysical properties – for example, QPI can measure a cell's dry mass by integrating refractive index differences across the cell. Such metrics will be analyzed to distinguish tumor cells from white blood cells and to assess features linked to metastatic potential. The consortium will also develop AI algorithms to analyze the QPI data and improve diagnostic accuracy. Machine learning models will be trained on the QPI-derived feature set to automatically identify CTCs and evaluate their malignancy. Notably, AI-powered image analysis has already proven highly effective at differentiating CTCs from background blood cells, which underscores its importance in our platform. The AI will correlate QPI biomarkers with patient outcomes to discover patterns predictive of metastasis. All software components will be implemented on a cloud-based Software-as-a-Service (SaaS) platform, allowing remote data processing and seamless integration into hospital workflows. Additionally, the project will initiate the preparation of CE-IVDR documentation to ensure the platform meets European regulatory requirements for in-vitro diagnostics. By the project's end, we aim to have a validated prototype system and the necessary documentation for CE-marking under the new IVDR, positioning the technology for clinical deployment.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Leads the design of the digital holographic QPI optical system and imaging methodology. Develops and optimizes the interferometric imaging hardware, including illumination and sensor configuration, to achieve high-resolution phase images of CTCs with high speed. Responsible for core algorithms (phase reconstruction, aberration compensation, and quantitative image analysis) to accurately compute cellular biomarkers from holograms. Guides the quality management and regulatory strategy by establishing a CE-IVDR compliant documentation framework. Provides validation support through international standardization efforts – e.g. contributing to reference measurement standards for QPI – and leverages its research network to ensure the technology meets global diagnostic standards. • Development and optimization of **DHM/QPI optical systems** for label-free CTC imaging • Phase reconstruction, aberration correction, and quantitative parameter extraction algorithms • Joint definition and validation of QPI-based **digital biomarkers** • Support for **CE-IVDR–aligned technical documentation** and optical system validation • Contribution to international standardization and joint IP generation
Expected R&R of Korean SME Partner	• **Lead organization** responsible for overall project coordination and execution • Provision of **clinically validated CTC isolation and analysis services** • Recruitment of clinical partners and management of IRB-approved clinical samples • Construction and curation of large-scale clinical QPI datasets • Validation of QPI biomarkers against clinical outcomes (metastasis, treatment response) • Global commercialization and SaaS deployment via **Japan and U.S. subsidiaries**

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **CE-ready QPI Diagnostic System:** A jointly developed hardware/software platform for label-free CTC detection, ready for CE-IVDR certification with documentation of safety, performance, and manufacturing process. • **AI-Powered Digital Biomarker Suite:** An advanced analytical software that uses AI to quantify QPI-derived biomarkers and provides clinicians with objective metrics for metastatic risk assessment. • **Validated CTC Detection Pipeline:** A fully tested workflow for blood-based CTC analysis in metastatic breast cancer, demonstrated in clinical pilot studies. The pipeline will show improved sensitivity and specificity in identifying CTCs and predicting metastasis, as benchmarked against current standards. • **Foundation for Global Commercialization:** Establishment of a Korea–Fraunhofer collaboration model (including potential joint venture) to bring the product to international markets. This includes agreements for co-ownership of intellectual property, a roadmap for large-scale manufacturing, and alignment with EU regulatory and standardization pathways for a smooth entry into the European healthcare market.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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